

Artificial Neural Network Applications

Wen-Ren Yang

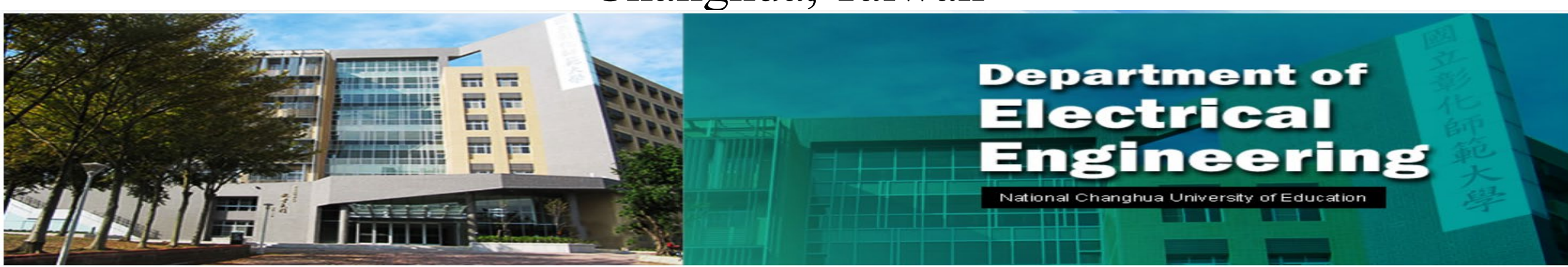
wry87c@cc.ncue.edu.tw, [linkedin.com/in/wry87c](https://www.linkedin.com/in/wry87c)

Associate Professor

Department of Electrical Engineering

National Changhua University of Education

Changhua, Taiwan



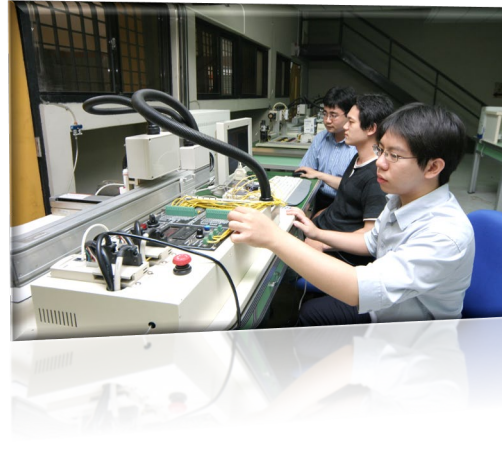
Internship Tasks

- Digital Signal Processing (DSP): short-time fourier transform, discrete wavelet transform, and digital filtering
- Artificial Intelligence and Machine Learning (AI-ML): long-short term memory and general regression neural networks
- Matlab codes, simulation, and analysis for DSP & AI-ML
- Python codes, simulation, and analysis for DSP & AI-ML
- DSP & AI-ML implementation on the embedded system (FPGA chipset)
- Weekly report and meeting.

Research Topics

- Smart Grid and Power Quality
 - (1) Feature Extraction
 - (2) System Response Identification
 - (3) System Status Identification
- Embedded Image Recognition Applications
 - (1) Feature Extraction
 - (2) Contour Identification
 - (3) Object Recognition

College of Engineering & Campus Life



Edge Computing & AI for NCUE BESS

